电池冷却系统
BTMS Test Instruction

测
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书

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一、电池冷却系统测试流程 BTMS testing process：

1、BTMS 安装固定后，清理 BTMS 上的所有杂物；

1. After installing and fixing the BTMS, clean all debris on the BTMS;

2、BTMS 抽真空，加注制冷剂；

2. Vacuum BTMS and add refrigerant;

3、BTMS 连接高低压线束；

3. BTMS connects high and low voltage harnesses;

4、BTMS 连接冷却管路部分，冷却管路抽真空加注冷却液；

4. Connect the cooling pipeline section of BTMS, vacuum the cooling pipeline and add coolant;

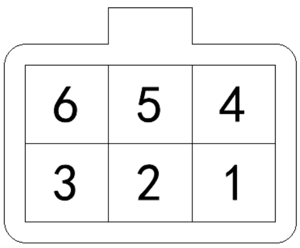
注：无关人员不要靠近 BTMS，避免发生危险：有关调试人员也禁止接触电气接线金属导电部分。

Note: Unrelated personnel should not approach the BTMS to avoid danger: commissioning personnel are also prohibited from touching the metal conductive parts of electrical wiring.

二、BTMS 调试面板电气接口定义 BTMS Control Panel Electrical Interface Definition：

1、调试面板端子接口定义；

1. Control panel terminal interface definition;



control panel
(Connector View)

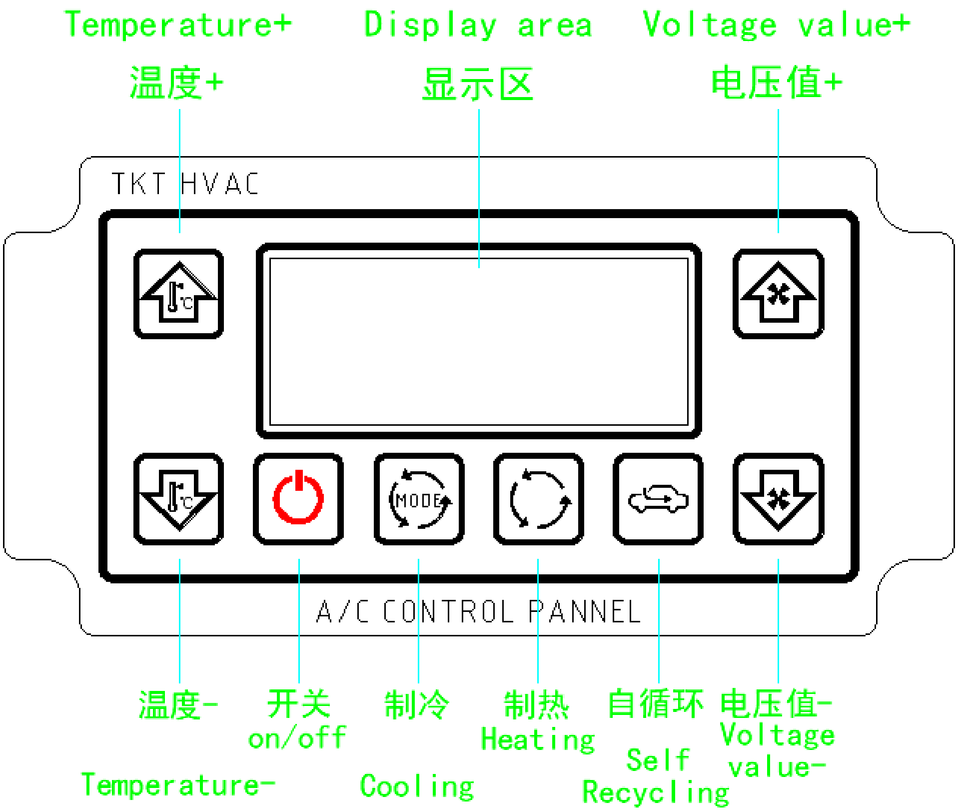
模拟车通信面板端子接口定义 Definition of Control Panel Terminal Interface

针脚号 Pin number	定义 Definition	电气特征 Electric characteristics	注释 Annotation
P1	CAN1-L		保留 Reserve
P2	CAN2-L	CAN-L（Connect Control Module）	CAN-L (Commissioning BTMS)
P3	GND	12/24V-	电源地 Power supply ground
P4	CAN1-H		保留 Reserve
P5	CAN2-H	CAN-H（Connect Control Module）	CAN-H (Commissioning BTMS)
P6	24V+	12/24V+	电源正 Power supply positive

三、BTMS 调试面板使用说明 Instructions for using BTMS control panel：

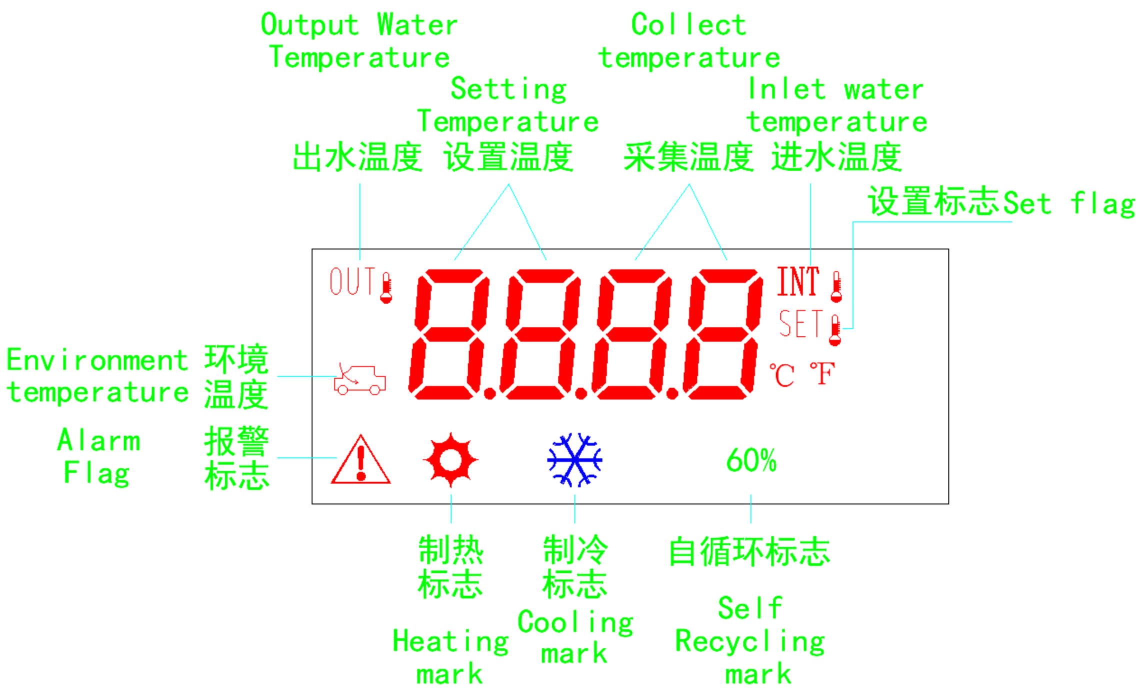
1、调试面板图示；

1. Control panel diagram;



2、调试面板图示；

2. Control panel display area diagram;



3、按键及显示功能说明；

3. Key and Display Function Description;

3.1 开/关机键 On/Off button：

短按开机，长按 1 秒关机。

Short press to turn on, long press for 1 second to turn off.

3.2 温度加 Temperature+：

短按，设定温度加 1° C。

Short press to increase the set temperature by 1 ° C.

长按，可切换 CAN 通信帧数据格式。图标“LOW”点亮,表示当前为 Inter 帧数据格式；图标“HIGH”点亮，表示当前为 Motorola 帧数据格式；

Long press to switch the CAN communication frame data format. The icon "LOW" lights up, indicating that it is currently in the Inter frame data format; The icon "HIGH" lights up, indicating that it is currently in Motorola frame data format.

3.3 温度减 Temperature-：

短按，设定温度减 1° C。

Short press to set the temperature to decrease by 1 ° C.

3.4 电压值加 Voltage value+：

长按，进入到模拟 BMS 发送当前电压值的界面；然后短按，为电压值加。

Long press to enter the interface for simulating BMS to send the current voltage value; Then short press to increase the voltage value.

3.5 电压值减 Voltage value-：

短按，切换“出水温度—进水温度—环境温度—故障代码”显示。

Short press to switch between "outlet temperature - inlet temperature - Environment temperature - error code"display.

如果长按电压值加键，进入到模拟 BMS 发送当前电压值的界面，然后短按电压值减，为电压值减。

If you long press the “voltage value +” button, you will enter the interface for simulating BMS to send the current voltage value, and then short press the “voltage value -” button to decrease the voltage value.

3.6 制冷键 Cooling ：

3.6.1 功能描述：制冷模式。Function description: Cooling mode.

3.6.2 图标提示：选定制冷模式时，图标“❄”闪烁表明当前是制冷等待，等待关闭上一个工作模式或者温度条件未达到；图标“❄”常亮则说明进入制冷模式。Icon prompt: When the cooling mode is selected, the flashing “❄” icon indicates that the current mode is cooling waiting, waiting to turn off the previous working mode, or the temperature condition has not been reached; If the icon “❄” stays on, it indicates entering cooling mode.

3.7 制热键 Heating：

3.7.1 功能描述：制热模式。Function description: Heating mode.

3.7.2 图标提示：选定制热模式时，图标“☼”闪烁表明当前是制热等待，等待关闭上一个工作模式或者温度条件未达到；图标“☼”常亮则说明进入制热模式。Icon prompt: When the heating mode is selected, the icon “☼” flashes to indicate that it is currently waiting for heating, waiting to close the previous working mode, or the temperature condition has not been met; If the icon “☼” stays on, it indicates that the heating mode has been entered.

3.8 自循环键 Self Recycling：

3.8.1 功能描述：自循环模式。Function description: Self Recycling Mode.

3.8.2 图标提示：选定自循环模式时，“60%”图标闪烁表明当前是自循环等待，等待关闭上一个工作模式；“60%”图标常亮则说明进入自循环模式。Icon prompt: When the auto cycle mode is selected, the“60%”icon flashes to indicate that it is currently auto cycle waiting, waiting to close the previous working mode; If the icon“60%”stays on, it indicates that the device has entered self cycling mode.

4、操作说明；

4. Operating instructions；

4.1 当热管理控制器产生故障时，则数码管“采集温度”区域显示报警值，图标“ ”点亮。When the thermal management controller malfunctions, the "Collect temperature" area of the digital tube displays an alarm value, and the icon " " lights up.

4.2 长按“电压值加”键，进入到模拟 BMS 发送当前电压值的界面，可通过“电压值加”和“电压值减”更改模拟 BMS 发送当前电压值。Long press the "Voltage Value +" button to enter the interface where the current voltage value is sent by the analog BMS. You can change the current voltage value sent by the analog BMS through "Voltage Value +" and "Voltage Value -".

当电压值无更改 8 秒后，显示界面恢复正常显示状态。After 8 seconds of no change in voltage value, the display interface returns to normal display state.

4.3 短按开/关机后，默认不发送任何模式给热管理控制器，可通过按键操作对热管理发送工作模式。例：短按制冷键，发送制冷命令给热管理控制器，通过温度加、减键设定目标温度。After short pressing the "On/Off" button, no mode will be sent to the thermal management controller by default. The working mode of the thermal management can be sent by pressing the button. Example: Short press the "cooling" button to send a cooling command to the thermal management controller, and set the target temperature through the "temperature+" and "temperature-" keys.

四、故障代码列表 Fault Code Table：

Description of fault alarm values

序号 S/N	故障代码 Error Code	故障内容 Fault Content	造成原因 Cause	故障等级 Fault Level
1		无故障 Trouble-free		
2	48	BMS 和 TMS “CAN” 通讯异常 BMS and TMS "CAN" communication abnormality	机组不受控制（判断时间需≤7s） The unit is out of control (judgment time needs to be ≤7s)	Level1
3	1	PTC 1 级故障 PTC level 1 fault	IGBT 故障/或 IGBT 击穿 IGBT failure/or IGBT breakdown	Level1
4	2	PTC 2 级故障 PTC level 2 fault	PTC 过压 PTC overvoltage	Level2
5	5	PTC 2 级故障 PTC level 2 fault	PTC 欠压 PTC undervoltage	Level2
6	11	PTC 2 级故障 PTC level 2 fault	NTC 坏/或 PTC 自身故障 NTC damaged/or PTC itself faulty	Level2

序号 S/N	故障代码 Error Code	故障内容 Fault Content	造成原因 Cause	故障等级 Fault Level
7	14	PTC 2 级故障 PTC level 2 fault	PTC 通讯故障（或生命帧停止更新时间大于 10 秒） PTC communication failure (or life frame stops updating for more than 10 seconds)	Level2
8	19	AC 2 级故障 AC level 2 fault	压缩机故障（制冷模式） Compressor failure (cooling mode)	Level2
9	20	BMS 高压供电故障 BMS high voltage power supply failure	在制冷或制热工作时需要使用高压设备(压缩机或 PTC)时,BMS 提供的电压过高或过低。 When high-voltage equipment (compressor or PTC) is required for cooling or heating work, the voltage provided by the BMS is too high or too low.	Level2
10	21	Fan 2 级故障 Fan level 2 fault	冷凝风扇故障电源故障 Condenser fan failure Power supply failure	Level2
11	22	高压传感器 2 级故障 High voltage sensor level 2 fault	压缩机高压过高，过低或者高压传感器开路，短路，系统压力过高（制冷模式） The compressor high pressure is too high or too low or the high pressure sensor is open or short circuited, and the system pressure is too high (cooling mode)	Level2
12	23	低压传感器 2 级故障 Low voltage sensor level 2 fault	压缩机低压过高，过低或者低压传感器开路，短路，系统压力过低（制冷模式） The compressor low pressure is too high or too low or the low pressure sensor is open or short circuited, and the system pressure is too low (cooling mode)	Level2
13	24	机组进水温度传感器 2 级故障 Unit inlet water temperature sensor Level 2 fault	机组进水温度传感器故障，开路， 短路（全模式） The unit inlet water temperature sensor is faulty, open circuit, short circuit (all modes)	Level2
14	25	机组出水温度传感器 2 级故障 Unit outlet water temperature sensor Level 2 fault	机组出水温度传感器故障，开路， 短路（全模式） The unit outlet water temperature sensor is faulty, open circuit, short circuit (all modes)	Level2
15	26	机组环境温度传感器 2 级故障 Unit ambient temperature sensor	机组环境温度传感器故障，开路， 短路（全模式） Unit ambient temperature sensor failure, open circuit, short circuit (all modes)	Level2

序号 S/N	故障代码 Error Code	故障内容 Fault Content	造成原因 Cause	故障等级 Fault Level
		Level 2 fault		
16	27	水泵 water pump	水泵电源故障 Water pump power failure	Level2
17	30	水位传感器 2 级故障 Water level sensor level 2 fault	水位传感器故障，开路或者短路（全模式） Water level sensor failure, open circuit or short circuit (all modes)	Level2
18	31	PTC 2 级故障 PTC Level 2 fault	PTC 出水温度高/或 PTC 过温故障 PTC outlet water temperature is high/or PTC over temperature fault	Level2
19	47	低压供电 2 级故障 Low voltage power supply level 2 fault	低压供电低于 18V 或者高于 32V Low voltage power supply is lower than 18V or higher than 32V	Level2
20	51	机组防冻温度传感器 2 级故障 Unit anti-freeze temperature sensor level 2 fault	机组防冻温度传感器故障，开路， 短路（全模式） Unit antifreeze temperature sensor failure, open circuit, short circuit (all modes)	Level2

注：如果设备为单冷型，则可以忽略制热部分功能及故障反馈。
Note: If the device is a single cooling type , the heating function and fault feedback can be ignored.

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